RAG Evaluation

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RAG Evaluation:

Possible Issues

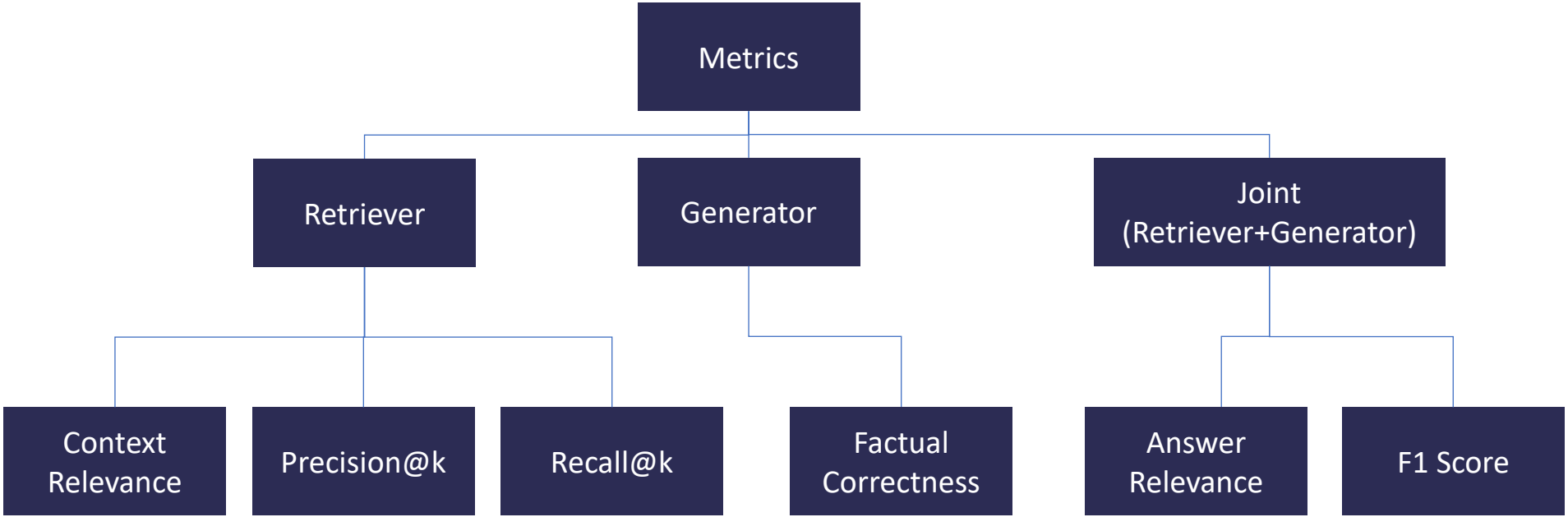
RAG Evaluation

Issues

- hallucinations due to
  - Retriever…fails to retrieve relevant context
  - Generation…LLM does not consider context
  - Generation…LLM picks irrelevant information from context

RAG Evaluation

Metrics



RAG Evaluation

Required Abilities

Noise Robustness

- possible that document is related to user query, but does not have relevant info

Negative Rejection

- LLMs always generate text
- ability to NOT give answer when no info is available

Information Integration

- ability to assimilate information from multiple docs

Counter Fact Robustness

- info might be wrong
- RAG should reject know inaccuracies

Latency

Bias

Toxicity

RAG Evaluation

Evaluation Metrics

Accuracy

$$= \frac{count\ relevant + irrelevant\ documents\ retrieved}{Total\ number\ docs}$$

Precision

$$= \frac{count\ relevant\ documents\ retrieved}{Total\ count\ docs}$$

Precision@k

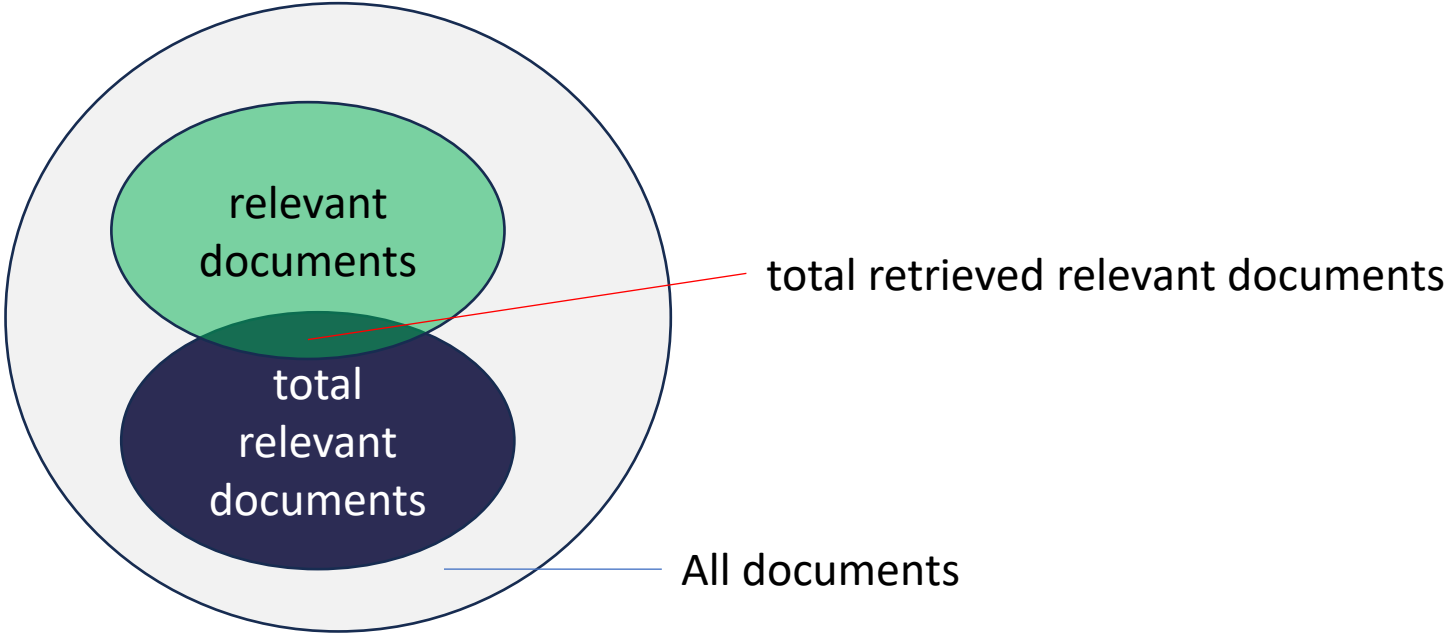
$$= \frac{count\ relevant\ documents\ retrieved\ in\ Top - k\ docs}{Top - k\ documents\ count}$$

Recall

$$= \frac{count\ relevant\ documents\ retrieved}{Count\ relevant\ documents\ count}$$

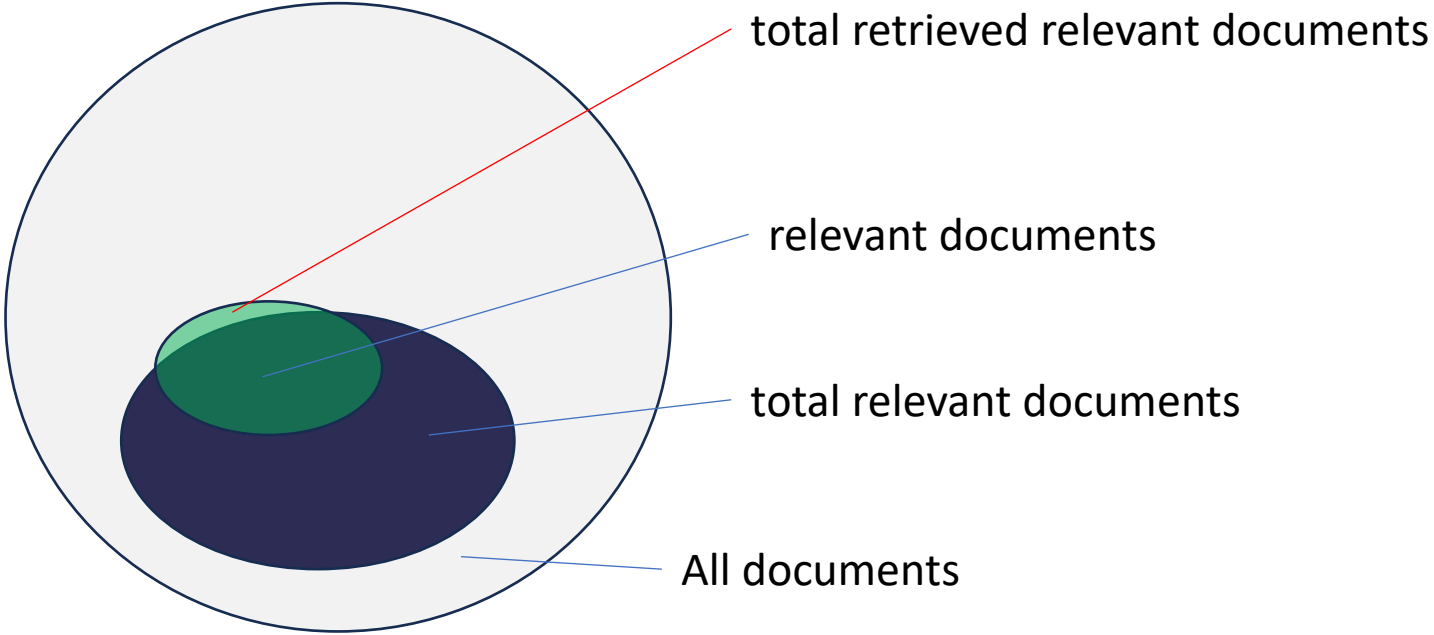
RAG Evaluation

Evaluation Metrics



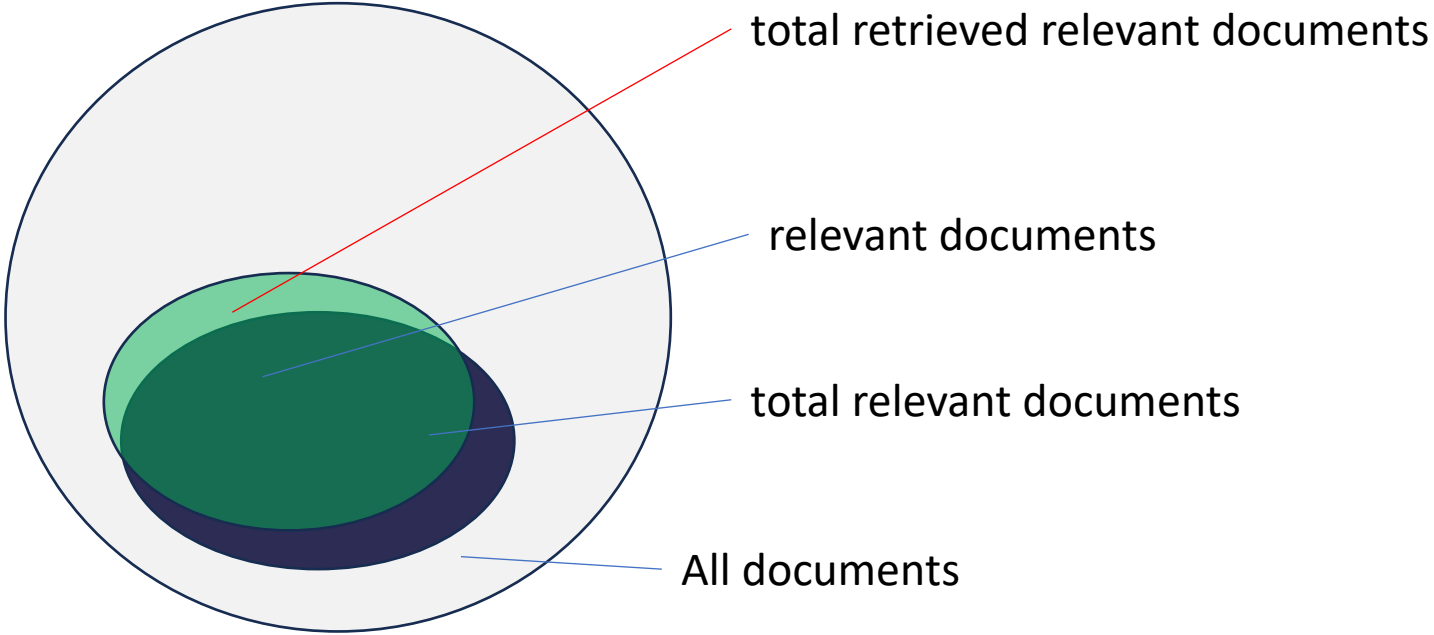
RAG Evaluation

High Precision, Low Recall



RAG Evaluation

High Precision, High Recall



RAG Evaluation

RAG Questions and KPIs: Context Relevance

Questions	KPIs	Definitions
Retrieval ▪ Is retrieval needed to answer user query?	Context Relevance	$= \frac{Count\ relevant\ sentences\ in\ retrieval}{Total\ count\ sentences\ in\ retrieval}$

RAG Evaluation

RAG Questions and KPIs: Context Relevance

Context

Apple Inc. is a multinational technology company headquartered in Cupertino, California.
The company designs, manufactures, and sells consumer electronics, software, and online services.
Its flagship products include the iPhone, iPad, and MacBook.
Recently, Apple announced the launch of its latest AI-powered wearable device.
In unrelated news, a global survey on renewable energy adoption showed significant growth in solar energy usage in Europe.

User Query

What are the latest AI-powered products launched by Apple?

Total count sentences = 5
Relevant sentences = 4
context relevance = 80%

RAG Evaluation

RAG Questions and KPIs: Answer Faithfulness

Questions	KPIs	Definitions
Generation ▪ Is answer based on retrieval?	Answer Faithfulness	$= \frac{Count\ generated\ claims\ in\ context}{Total\ count\ claims\ in\ response}$

RAG Evaluation

RAG Questions and KPIs: Answer Faithfulness

Context

The 2020 United States presidential election was held on November 3, 2020. Joe Biden, the Democratic candidate, defeated the incumbent president, Donald Trump, to become the 46th president of the United States. Biden's victory was confirmed by the Electoral College on December 14, 2020.

User Query

Who won the last U.S. presidential election in 2020, and when exactly was it held?

Joe Biden won the U.S. presidential election which was held on November 8, 2020.

partially faithful answer

RAG Evaluation

RAG Questions and KPIs: Answer Relevance

Questions	KPIs	Definitions

Generation

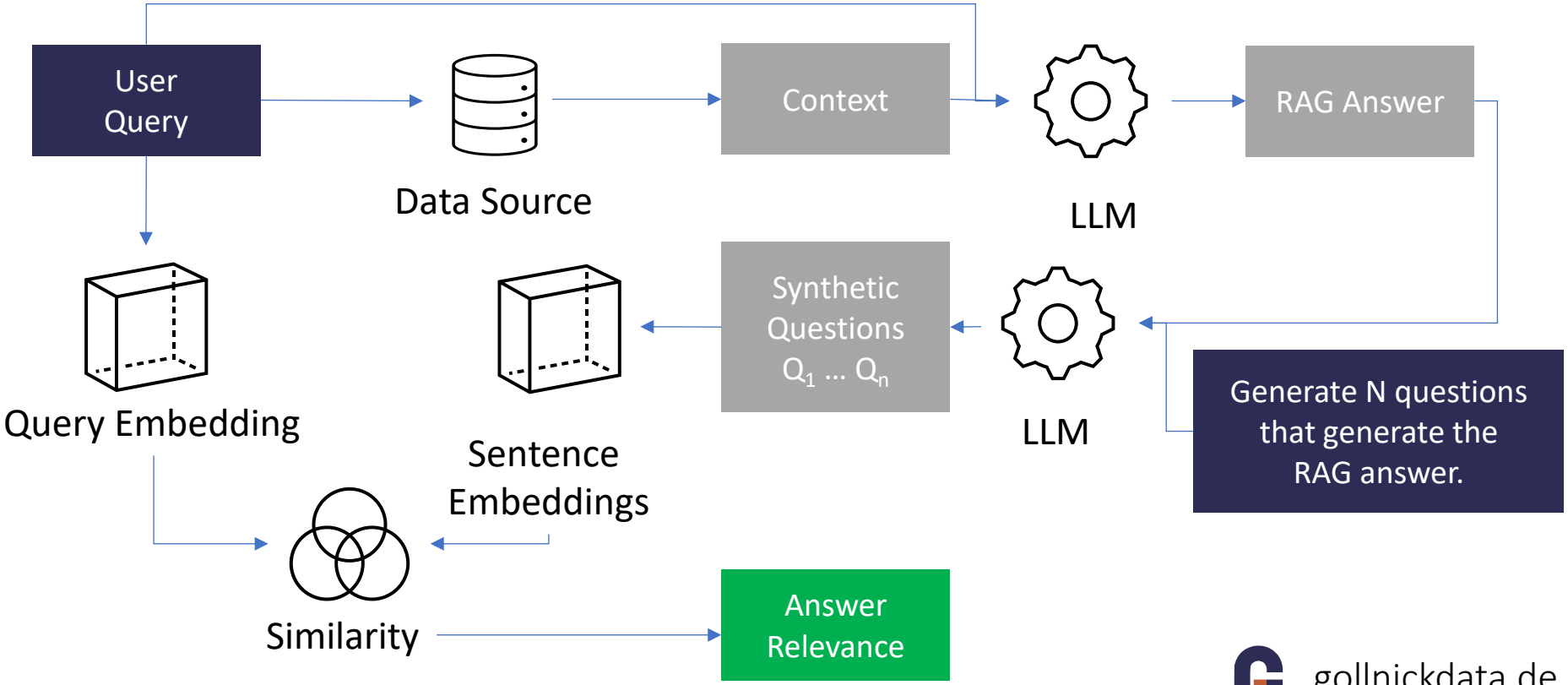
- Is answer relevant to the user query?

Answer Relevance

$$= \sum_{i=0}^{N} Similarity(user\ query, LLM generated\ synthetic\ query[i])$$

RAG Evaluation

RAG Questions and KPIs: Answer Relevance



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